



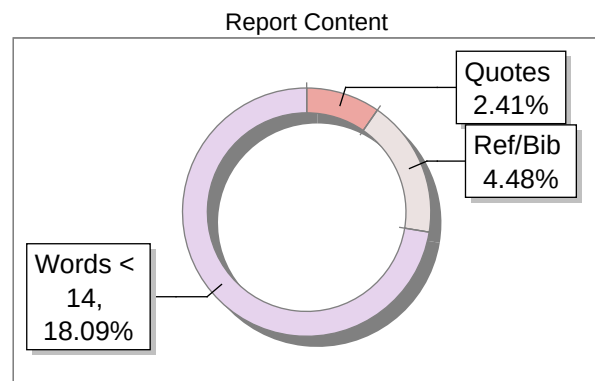
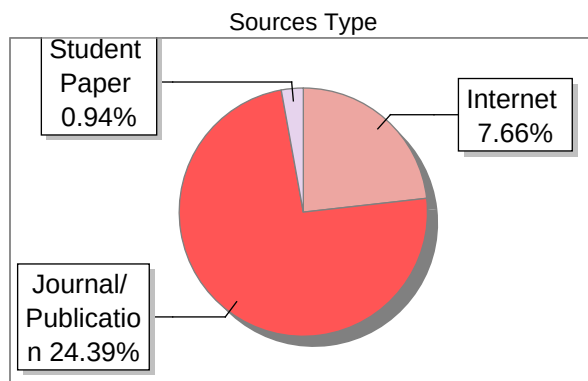
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Machine Learning and Deep Learning-Based Intrusion Detection Systems for IoT Security: Trends, Challenges, and Future Directions

Kalyan Sharma¹

Research Scholar, Department of Computer Science, University of Kota, Rajasthan, India

K K Sharma²

Associate Professor, Department of Computer Science, University of Kota, Rajasthan, India

Abstract

The rapid expansion of Internet of Things (IoT) networks has significantly increased cybersecurity risks due to the large-scale deployment of interconnected, resource-constrained, and heterogeneous devices. Traditional security mechanisms and signature-based Intrusion Detection Systems (IDS) are no longer sufficient to detect modern sophisticated attacks such as Distributed Denial of Service (DDoS), botnets, spoofing, and zero-day attacks [1,2]. To address these challenges, Artificial Intelligence (AI), including Machine Learning (ML) and Deep Learning (DL), has emerged as a powerful approach for intrusion detection in IoT environments [3,8].

AI-based IDS systems enable automatic learning from network traffic data, allowing real-time detection of anomalies and malicious activities with improved accuracy and reduced false positives. This paper provides a comprehensive survey of AI-based intrusion detection systems for IoT networks from 2020 to 2026. It discusses system architecture, detection techniques, IDS types, datasets, advantages, limitations, and future research directions. The study concludes that AI-based IDS frameworks are essential for securing next-generation IoT ecosystems.

Keywords: IoT Security, Intrusion Detection System, Artificial Intelligence, Machine Learning, Deep Learning, Cybersecurity, Edge Computing, Federated Learning, Anomaly Detection

1. INTRODUCTION

The Internet of Things (IoT) represents a revolutionary paradigm in modern computing where billions of physical devices are interconnected through the internet to collect, process, and exchange data [2]. These devices include smart sensors, wearable devices, industrial machines, healthcare monitoring systems, smart home

appliances, autonomous vehicles, and smart city infrastructure. The continuous growth of IoT has

significantly transformed industries by enabling automation, real-time decision-making, and intelligent data-driven services [2,8].

However, despite its advantages, IoT introduces severe cybersecurity challenges [1]. Unlike traditional computing systems, IoT devices are typically resource-constrained, meaning they operate with limited processing power, memory, storage capacity, and energy [1,2]. These limitations prevent the deployment of strong security mechanisms such as advanced encryption, continuous monitoring, and complex authentication protocols. As a result, IoT environments are highly vulnerable to cyberattacks [2].

Another major issue in IoT networks is their heterogeneous and distributed nature [6]. IoT systems consist of devices manufactured by different vendors, using different communication protocols, operating systems, and network architectures [6]. This lack of standardization creates security gaps that attackers can easily exploit. Additionally, many IoT devices are deployed in open or physically unsecured environments, making them more susceptible to tampering and unauthorized access [6].

Cybersecurity threats targeting IoT systems have increased rapidly in recent years [1]. Common attacks include Distributed Denial of Service (DDoS), botnet attacks (such as Mirai), spoofing, ransomware, data injection, and man-in-the-middle attacks [1,7]. These attacks can lead to service disruption, data loss, privacy breaches, and even physical harm in critical systems such as healthcare and industrial automation [4].

Traditional Intrusion Detection Systems (IDS), which rely on signature-based or rule-based methods, are no longer sufficient in modern IoT environments [2,6]. These systems can only detect previously known attacks and fail to identify new, evolving, or zero-day threats. Furthermore, they struggle to handle the massive volume and high-speed data generated by IoT networks [8].

To overcome these limitations, Artificial Intelligence (AI) techniques have emerged as a powerful solution [8]. Machine Learning (ML) and Deep Learning (DL) algorithms enable systems to learn patterns from network traffic data and automatically detect anomalies without predefined rules [3,8]. AI-based IDS can analyze large datasets in real time, classify network behavior, and detect both known and unknown attacks with high accuracy [3].

Between 2020 and 2026, significant advancements have been made in AI-driven IDS research [3,4,5]. Researchers have focused on improving detection accuracy, reducing false positives, and designing lightweight models suitable for edge and IoT devices [5]. Emerging technologies such as federated learning, explainable AI (XAI), and edge computing have further enhanced the effectiveness and scalability of IDS solutions [5,6].

Therefore, AI-based Intrusion Detection Systems are considered a critical component in securing next-generation IoT ecosystems, ensuring reliability, privacy, and resilience against evolving cyber threats [3,6].

II. RELATED RESEARCH WORK

Research on Intrusion Detection Systems for IoT networks has evolved significantly between 2020 and 2026, transitioning from traditional machine learning approaches to advanced deep learning and distributed AI-based frameworks [3,4].

In the early phase (2020–2021), researchers primarily focused on applying classical Machine Learning algorithms such as Support Vector Machines (SVM), Decision Trees, K-Nearest Neighbors (KNN), and Random Forest for intrusion detection [1,8]. These models demonstrated improved performance compared to traditional signature-based IDS. However, they still required manual feature engineering and struggled with high-dimensional IoT data [3].

During the same period, several studies highlighted the importance of feature selection and data preprocessing techniques to improve classification accuracy. Researchers also began exploring hybrid models combining multiple machine learning algorithms to enhance detection performance and reduce false alarms [1].

Between 2022 and 2023, the focus shifted towards Deep Learning-based IDS due to their ability to automatically extract meaningful features from raw network traffic [3,7]. Convolutional Neural Networks (CNN), Recurrent Neural Networks (RNN), and Long Short-Term Memory (LSTM) models became widely adopted for analyzing sequential IoT traffic data [3,7]. These models showed significant improvements in detecting complex and stealthy attacks such as botnets and advanced persistent threats (APTs) [3].

In addition, ensemble learning methods and hybrid architectures combining ML and DL were introduced to further improve robustness and generalization capability. Researchers also began addressing issues such as dataset imbalance using techniques like oversampling, undersampling, and synthetic data generation [3].

From 2024 onwards, research trends have shifted toward more advanced and distributed AI frameworks [4,5]. Federated Learning (FL) has gained significant attention as it allows multiple IoT devices to collaboratively train models without sharing raw data, thereby preserving privacy and reducing communication overhead [4]. This approach is particularly suitable for sensitive applications such as healthcare and smart homes [4].

Another emerging trend is Edge-Based Intrusion Detection Systems, where AI models are deployed closer to IoT devices to enable real-time detection with low latency [5]. This reduces dependency on cloud infrastructure and improves response time for critical applications.

Additionally, Explainable AI (XAI) has been introduced to address the interpretability issue in AI-based IDS [6]. Since deep learning models are often considered “black boxes,” XAI techniques help in understanding why a particular decision was made, increasing trust and reliability in security systems [6].

Recent surveys (2025–2026) indicate that although AI-based IDS systems significantly outperform traditional approaches in terms of accuracy and adaptability, several challenges remain unresolved. These include adversarial attacks on AI models, lack of standardized datasets, computational constraints in IoT devices, and difficulties in deploying real-time IDS in large-scale networks.

Overall, the literature clearly shows a continuous evolution from simple machine learning models to complex, intelligent, and distributed AI-based intrusion detection systems designed specifically for the dynamic nature of IoT environments [3,6].

III. SYSTEM ARCHITECTURE OF AI-BASED IDS

A. Overview

6 The system architecture of an AI-based Intrusion Detection System (IDS) for IoT networks is designed to efficiently monitor, analyze, and secure data flowing between interconnected IoT devices [2,6]. Unlike traditional IDS architectures, AI-based IDS integrates machine learning and deep learning models to enable intelligent decision-making, real-time anomaly detection, and adaptive learning capabilities [3,5].

IoT environments are highly distributed, heterogeneous, and resource-constrained; therefore, the IDS architecture must be lightweight, scalable, and capable of handling continuous high-speed data streams [5]. The architecture is typically divided into multiple functional layers, each responsible for a specific stage of intrusion detection, ranging from raw data collection to automated response [6].

B. Major Components of AI-Based IDS Architecture

1. Data Collection Layer

The data collection layer is the foundation of the IDS architecture [2]. It gathers raw data from various IoT devices, sensors, gateways, and network communication channels. The collected data includes:

- Network traffic packets (TCP/UDP flows) [1]
- System logs from IoT devices [4]
- Sensor-generated data (temperature, motion, etc.) [2]
- Application-level logs [1]

40 Since IoT devices generate continuous data streams, this layer must operate in real time and ensure minimal packet loss [8]. In many architectures, lightweight agents or edge gateways are used to collect and forward data to the preprocessing module [5].

2. Data Preprocessing Layer

Raw IoT data is often noisy, incomplete, and inconsistent [5]. The preprocessing layer transforms this raw data into a structured format suitable for AI processing. Key operations include:

- Data cleaning (removal of duplicate or corrupted records) [1]
- Handling missing values [3]
- Normalization and scaling of numerical features [8]
- Encoding categorical variables [1]
- Data transformation into feature vectors [3]

This step is crucial because the performance of AI models depends heavily on the quality of input data [3,8].

3 Feature Extraction and Selection Layer

IoT datasets often contain a large number of features, many of which may be irrelevant or redundant [3]. This layer selects the most significant features to improve detection accuracy and reduce computational complexity.

Common techniques include:

- Principal Component Analysis (PCA) [8]
- Information Gain [1]
- Chi-square test [3]
- Correlation-based feature selection [8]

26 Feature selection helps reduce overfitting and enhances the efficiency of machine learning models, especially in resource-constrained IoT environments [3].

4. AI-Based Detection Engine

This is the most critical component of the architecture. It uses Artificial Intelligence algorithms to classify network behavior into normal or malicious categories.

The detection engine may include:

- Machine Learning Models
- Random Forest
- Decision Trees
- Support Vector Machines (SVM)
- K-Nearest Neighbors (KNN)
- Deep Learning Models
- Artificial Neural Networks (ANN)
- 23 Convolutional Neural Networks (CNN)
- Long Short-Term Memory (LSTM) networks
- Hybrid Models

27 Combination of ML and DL techniques for improved accuracy and robustness.

47 The AI engine continuously learns from new data, enabling adaptive detection of evolving cyber threats such as DDoS attacks, botnets, spoofing, and zero-day attacks.

5. Decision-Making Module

After classification, the system determines whether the detected activity is:

- Normal behavior
- Suspicious behavior
- Malicious attack

20 If an anomaly is detected, the system assigns a risk score based on severity levels. This helps prioritize responses for critical threats [6].

6. Alert and Response System

Once an intrusion is detected, this module performs automated actions such as:

- Generating real-time alerts to administrators [1]
- Blocking suspicious IP addresses [6]
- Isolating compromised IoT devices [4]
- Logging attack details for forensic analysis [1]

In advanced systems, automated mitigation strategies are implemented using Software-Defined Networking (SDN) or cloud-based security controllers [5].

7. Cloud / Edge Integration Layer

Modern AI-based IDS architectures integrate cloud and edge computing:

- Edge computing enables real-time detection with low latency near IoT devices [5].
- Cloud computing provides high processing power for large-scale data analysis and model training [5,6].
- This hybrid approach ensures both scalability and real-time performance [5].

C. SYSTEM ARCHITECTURE DIAGRAM

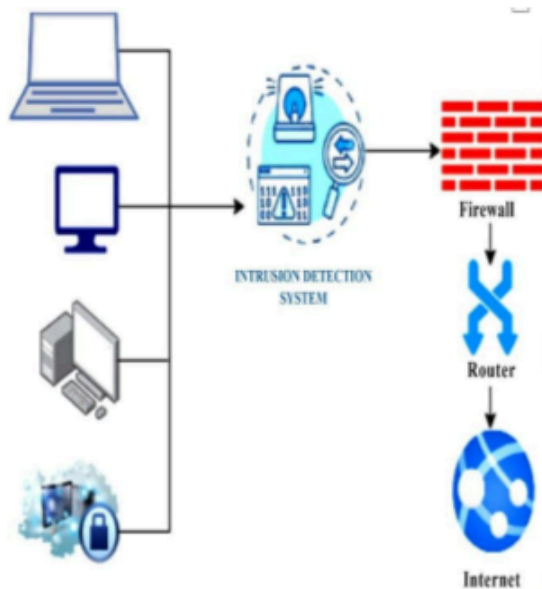


Figure 1: Use of Machine Learning and Deep Learning in Intrusion Detection for IoT

D. WORKING OF THE ARCHITECTURE

The working process of the AI-based IDS architecture begins when IoT devices continuously generate data [2]. This data is transmitted to the collection layer, where it is captured and forwarded to preprocessing modules. After cleaning and transformation, relevant features are extracted and passed to the AI detection engine [3].

The AI model analyzes patterns in the data and classifies behavior as normal or malicious. If an attack is detected, the decision-making module evaluates its severity and triggers the alert system. Finally, appropriate response actions are executed, such as blocking traffic or isolating compromised devices [4].

This entire process occurs in real time, enabling fast detection and response to cyber threats in IoT environments [5].

E. Importance of Architecture in IoT Security

The architecture plays a vital role in ensuring:

- Real-time intrusion detection
- Scalability for large IoT networks
- Efficient handling of high-speed data streams
- Reduced false positives using AI models
- Integration of edge and cloud computing

A well-designed AI-based IDS architecture ensures robust protection against modern cyber threats while maintaining system performance and reliability [4,6].

Bottom of Form

IV. AI TECHNIQUES USED IN IDS

Techniques used in ids:

A. Machine Learning Techniques

- Decision Tree
- Random Forest
- Support Vector Machine (SVM)
- K-Nearest Neighbors (KNN)
- Naïve Bayes

B. Deep Learning Techniques

- Artificial Neural Networks (ANN)
- Convolutional Neural Networks (CNN)
- Recurrent Neural Networks (RNN)
- Long Short-Term Memory (LSTM)

C. Unsupervised Learning

Used for detecting unknown attacks using clustering and anomaly detection techniques.

D. Federated Learning (2023–2026 Trend)

Enables distributed training across IoT devices without sharing raw data [4,5].

V. ADVANTAGES OF AI-BASED IDS

- **High detection accuracy:** Dramatically outpaces standard signature matchers on modern high-speed structures [3].
- **Real-time intrusion detection:** Recognizes exploitative behaviors inline as packets arrive at localized interfaces [5].
- **Ability to detect unknown attacks:** Catches complex zero-day threat patterns based on deviations from learned profiles [6].
- **Reduced manual monitoring:** Automated classification decreases constant checking requirements for analysts [1].
- **Scalability for large IoT networks:** Spreads calculation efforts across cloud-to-edge pathways seamlessly [5].
- **Adaptive learning capability:** Dynamically recalculates inner parameter weights as network habits modify [3].

VI. LIMITATIONS AND CHALLENGES

- **Limited computational resources in IoT devices:** Complex deep structures overtax limited processing cores [5, 6].
- **High false positive rates:** Slight configuration changes can mistakenly trigger priority alerts [1, 3].
- **Data imbalance issues:** Normal background traffic heavily outweighs rare historical exploit instances [3].
- **Privacy concerns in data sharing:** Uploading raw corporate network logs introduces vulnerability to metadata exposure [4].
- **Adversarial attacks on AI models:** Intentionally altered payloads can obscure the detection maps of models [3, 6].
- **Difficulty in real-time processing:** Immense scale demands can lead to hardware exhaustion and frame losses [5].

VII. AI BASED APPLICATIONS

AI-based IDS is widely used in:

A. Smart Homes

- Protects smart appliances, cameras, and IoT devices [2,4].

B. Healthcare Systems

- Secures patient data and medical IoT devices [4,5].

C. Industrial IoT (IIoT)

- Protects manufacturing and industrial control systems [6,7].

D. Smart Cities

- Secures traffic systems, surveillance, and utilities [4,6].

E. Autonomous Vehicles

- Ensures secure communication between vehicles and infrastructure [4,5].

VIII. FUTURE SCOPE

The future of AI-Based Intrusion Detection Systems (IDS) for IoT networks is highly promising due to the continuous growth of interconnected smart devices and the increasing complexity of cyber threats [4,6]. As IoT ecosystems expand across critical domains such as healthcare, smart cities, industrial automation, autonomous vehicles, and military applications, the demand for intelligent, adaptive, and real-time security solutions will continue to rise [4].

One of the most significant future directions is the development of lightweight AI models specifically designed for resource-constrained IoT devices [3]. Since most IoT nodes have limited processing power and memory, traditional deep learning models cannot be directly deployed [5]. Therefore, researchers are focusing on model compression techniques such as pruning, quantization, and knowledge distillation to enable efficient on-device intrusion detection [3].

Another important advancement is Edge AI-based IDS, where intrusion detection is performed at edge devices such as gateways and routers instead of centralized cloud servers. This approach reduces latency, improves response time, and enhances real-time threat detection capabilities, making it highly suitable for time-sensitive applications like autonomous vehicles and healthcare monitoring systems [5].

In addition, Federated Learning (FL) is emerging as a powerful approach for privacy-preserving IDS [4]. In FL, multiple IoT devices collaboratively train a shared global model without transferring raw data to a central server.

This ensures data privacy, reduces communication overhead, and improves scalability in large IoT networks.

Future IDS systems will also incorporate ²Explainable Artificial Intelligence (XAI) to ²⁴improve transparency and trust in security decisions [6]. ²⁴Since deep learning models are often considered “black boxes,” XAI techniques will help security analysts understand why a specific activity was classified as malicious, improving system reliability [6].

Furthermore, integration with Blockchain technology is expected to enhance data integrity and prevent tampering in IDS frameworks. Combined with real-time adaptive learning systems, future IDS will be capable of self-¹updating and responding dynamically to evolving cyber threats such as zero-day attacks and advanced persistent threats (APTs) [4].

Overall, the future scope of AI-based IDS lies in building intelligent, autonomous, scalable, and privacy-preserving security frameworks that can effectively secure next-generation IoT ecosystems [3,4].

IX. CONCLUSION

AI-based Intrusion Detection Systems ⁹represent a significant advancement in securing IoT networks. By leveraging machine learning and deep ¹⁴learning techniques, these systems can effectively ⁷detect both known and unknown cyber threats in real time. However, challenges such as resource constraints, ⁹privacy concerns, and evolving attack strategies must be addressed to fully realize their potential. Future advancements in edge computing and federated learning will further enhance IoT security.

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